Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative mortality (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 372 site-years) · deadtrees v2-2

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPP (g C m⁻² season⁻¹)

59% improved | 66% improved | 68% improved | 61% improved

NEP (g C m⁻² season⁻¹)

62% improved | 60% improved | 66% improved | 65% improved

ET (mm season⁻¹)

49% improved | 47% improved | 49% improved | 52% improved

WUE (g C mm⁻¹)

60% improved | 59% improved | 58% improved | 55% improved

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative mortality (%)
site maximum, 500 m
breaks 18.3 | 30.0

60
40
20
0